Lesson 03 Demo 05

Creating a New React Component Using useState Hook

Objective: To create a React component demonstrating the use of the useState hook, using the Vite framework for improved performance and fast development

Tools required: Visual Studio Code and Node.js

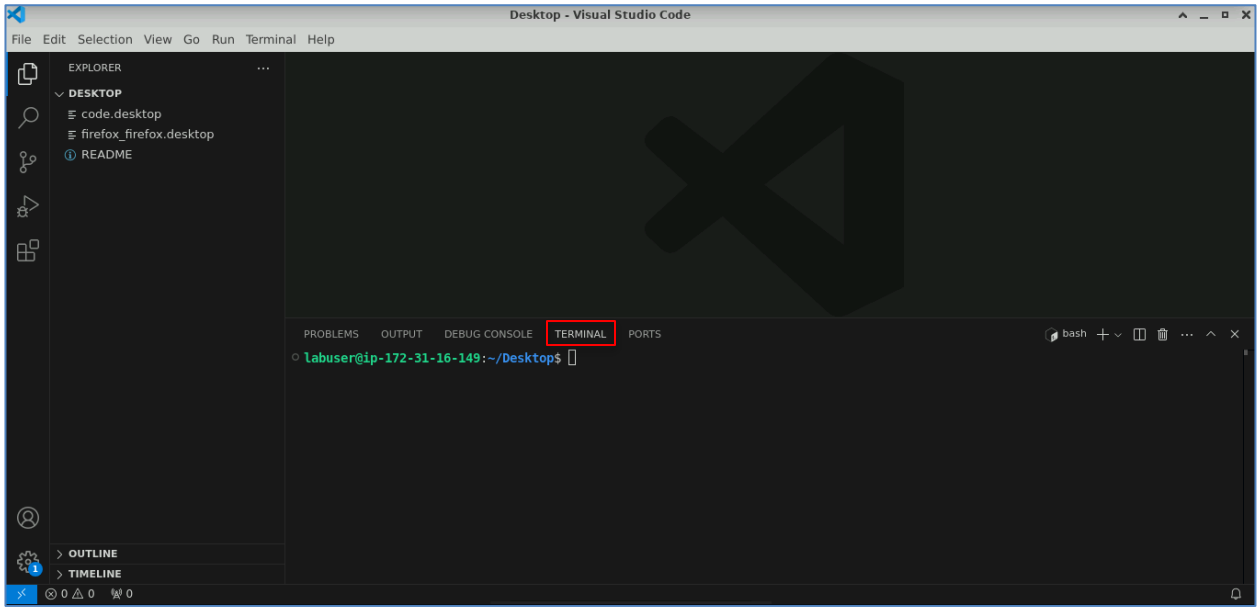
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a React component
3. Import and use the component
4. Run and verify the application

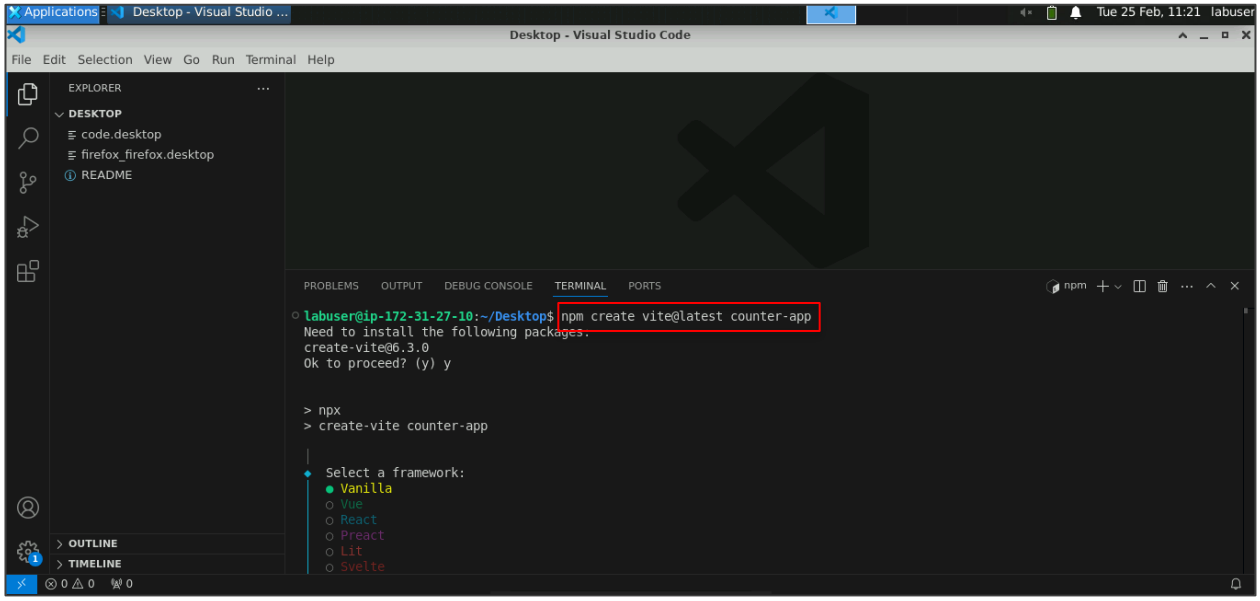
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

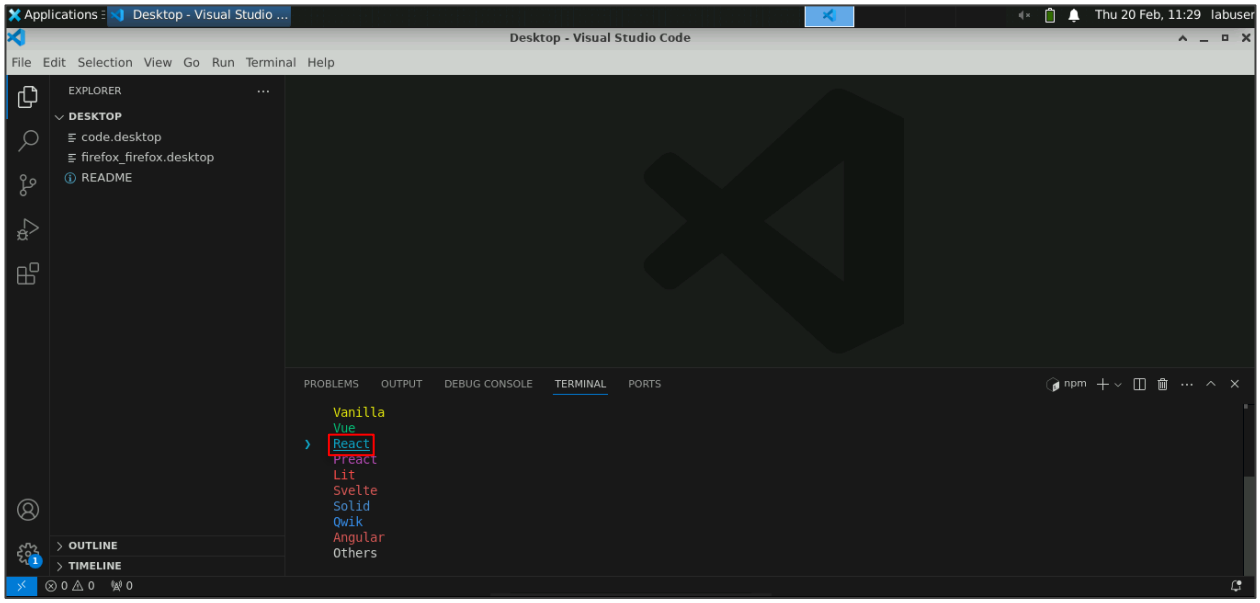


1.2 Run the following command to create a new React project using Vite:

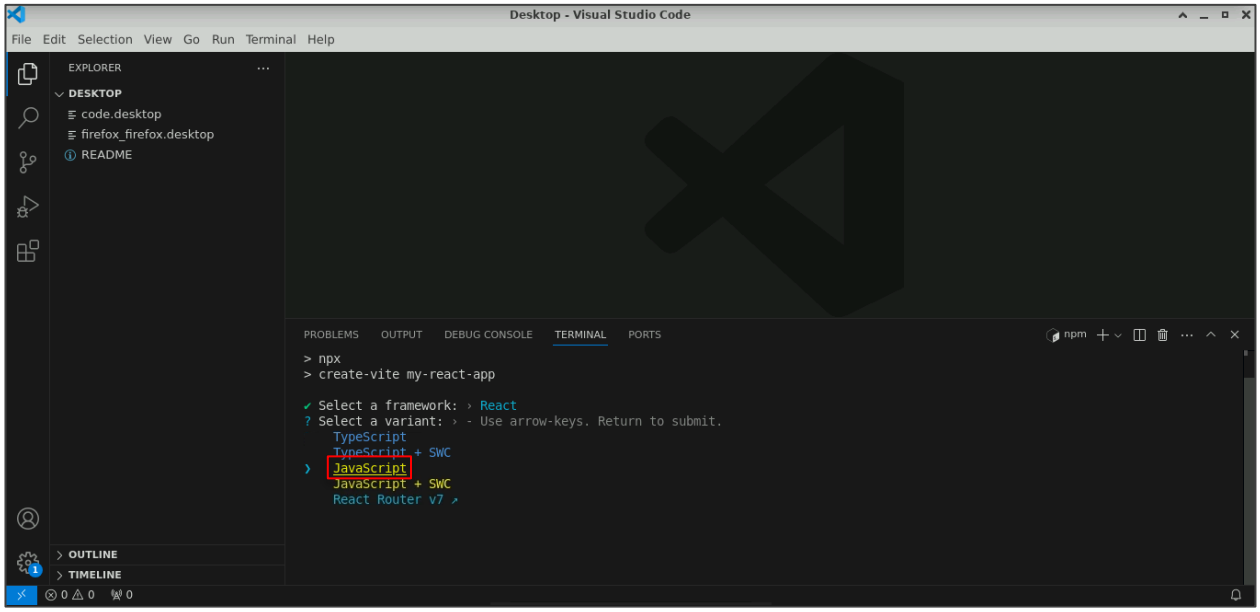
npm create vite@latest counter-app



1.3 Select **React** and press enter

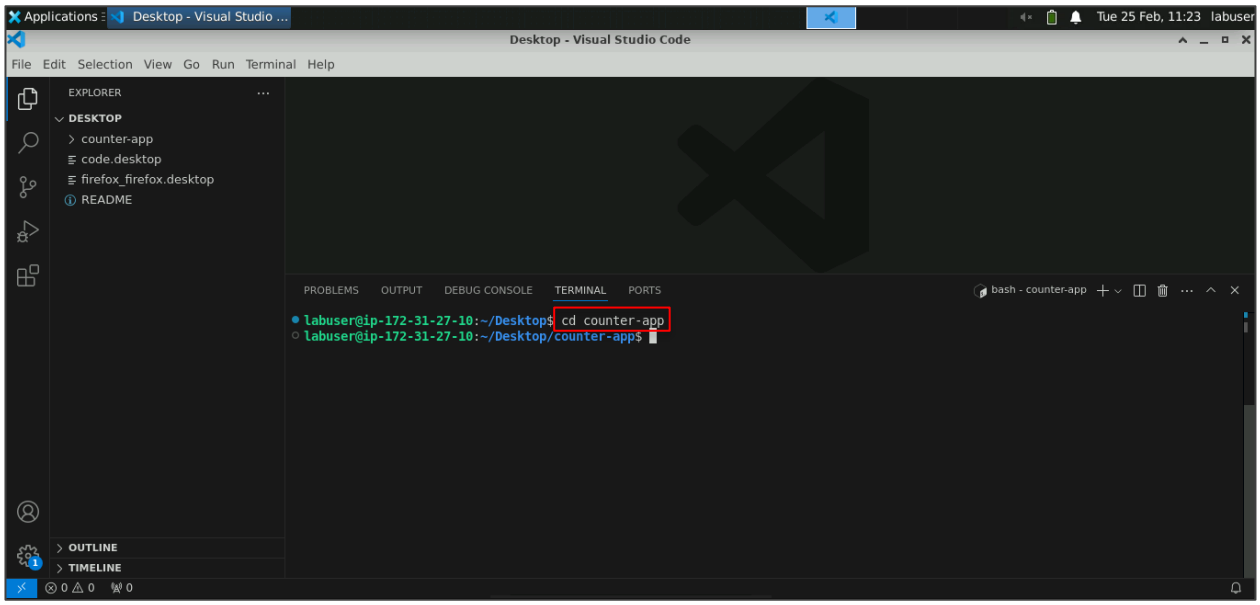


1.4 Select **JavaScript** and press enter



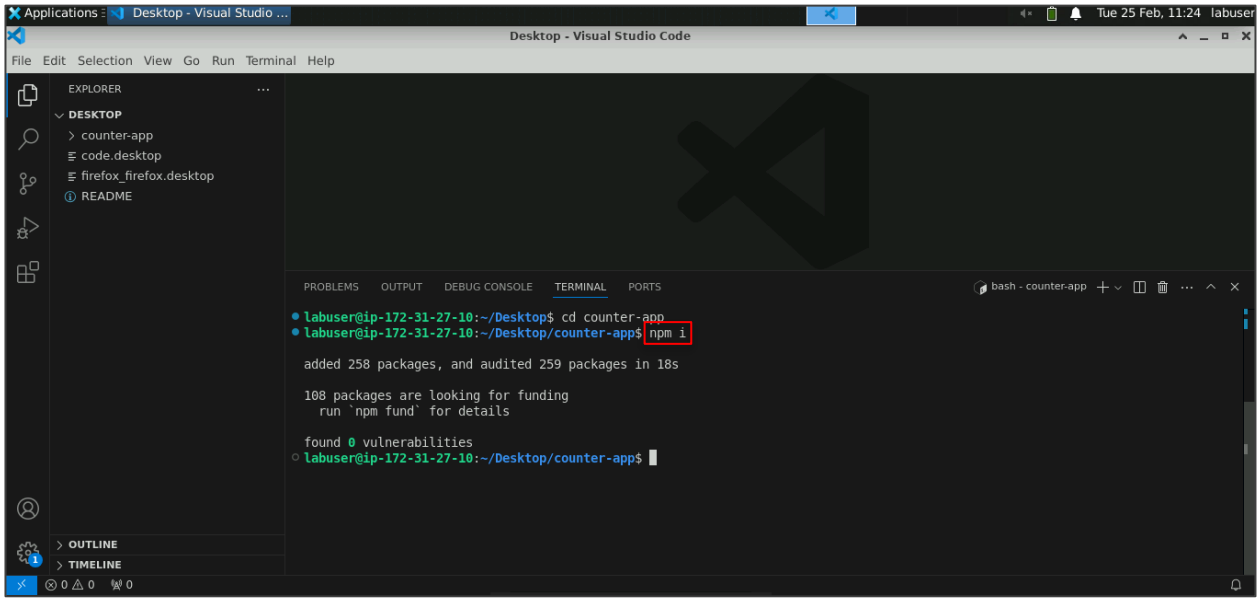
1.5 Run the following command to navigate to the project folder:

cd counter-app



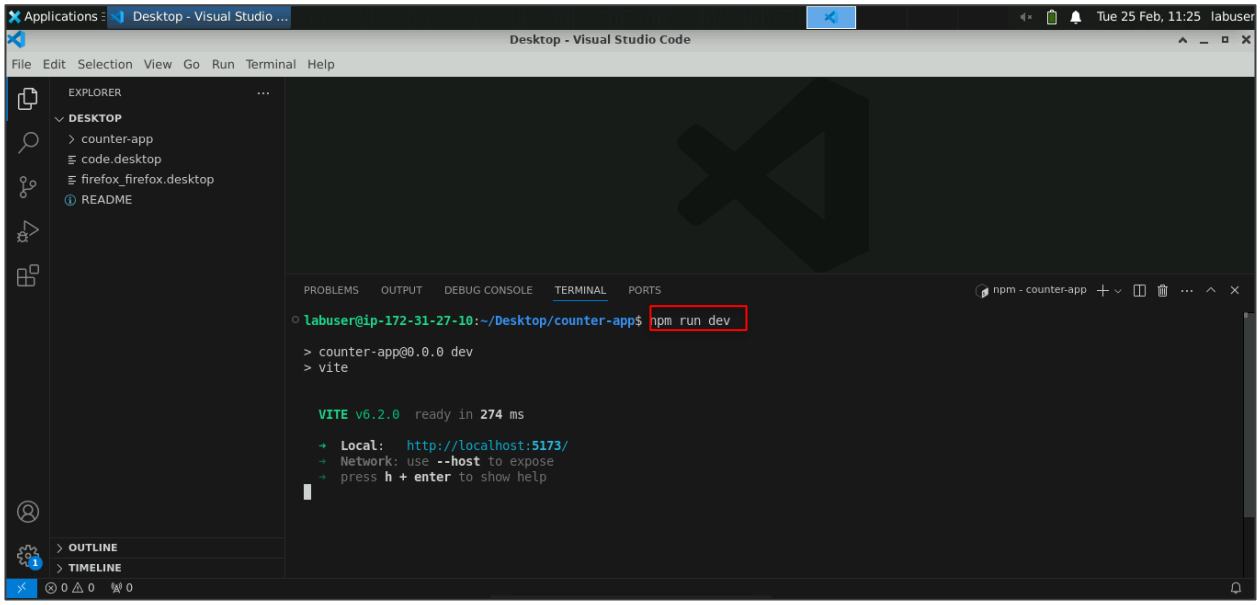
1.6 Run the following command to install dependencies:

npm i



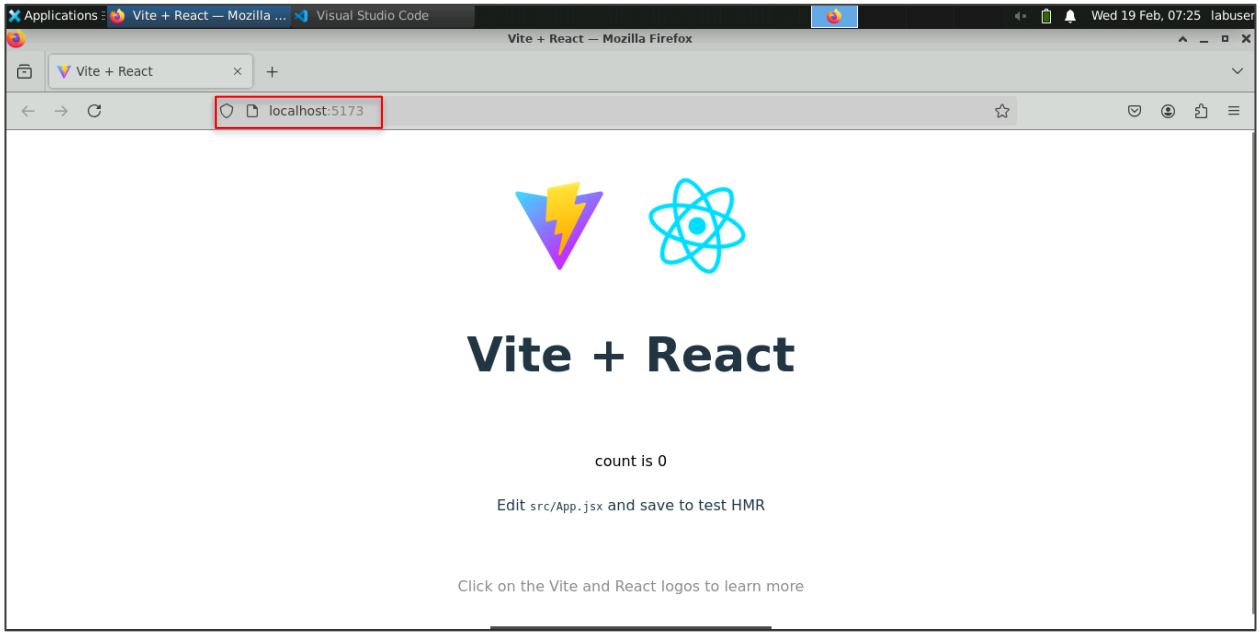
1.7 Run the following command to start the development server:

npm run dev



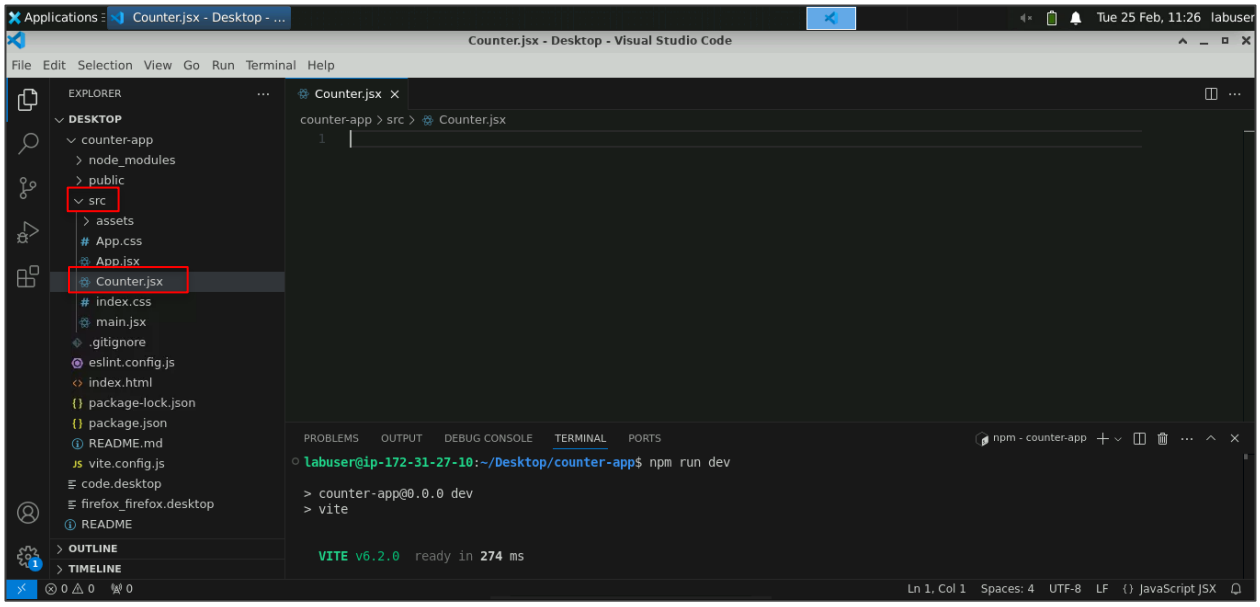
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



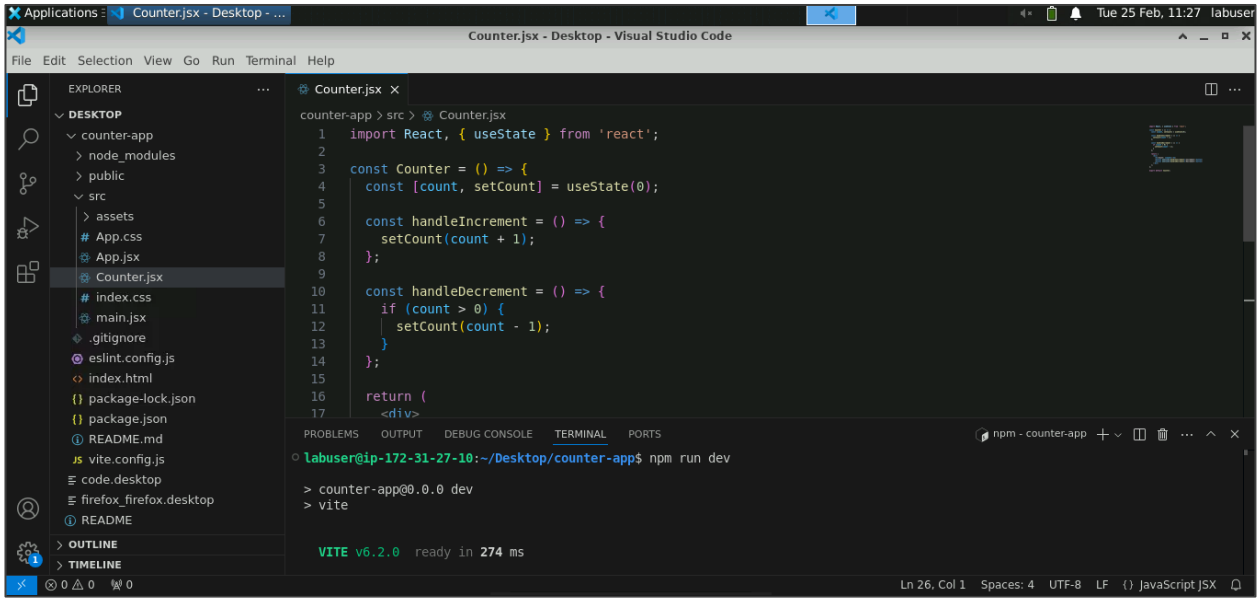
Step 2: Create a React component

2.1 Inside the **src** folder, create a new file named **Counter.jsx**



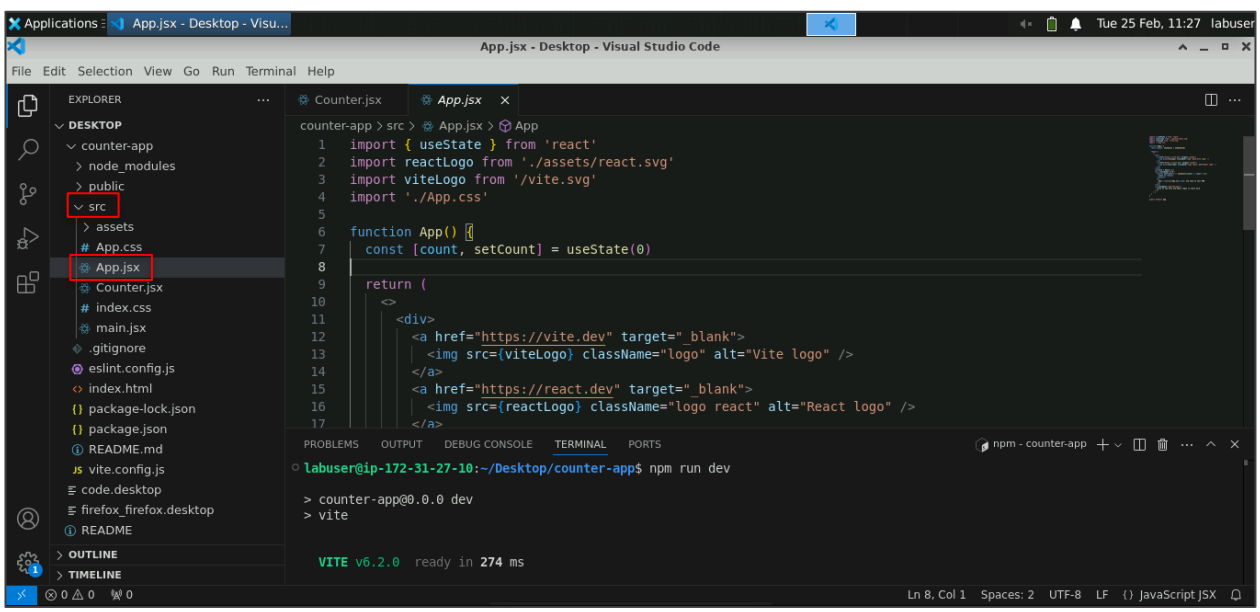
2.2 Add the following code to **Counter.jsx**

```
import React, { useState } from 'react';

const Counter = () => {
  const [count, setCount] = useState(0);

  const handleIncrement = () => {
    setCount(count + 1);
  };

  const handleDecrement = () => {
    if (count > 0) {
      setCount(count - 1);
    }
  };

  return (
    <div>
      <h2>Count: {count}</h2>
      <button onClick={handleIncrement}>Increment</button>
      <button onClick={handleDecrement}>Decrement</button>
    </div>
  );
};

export default Counter;
```



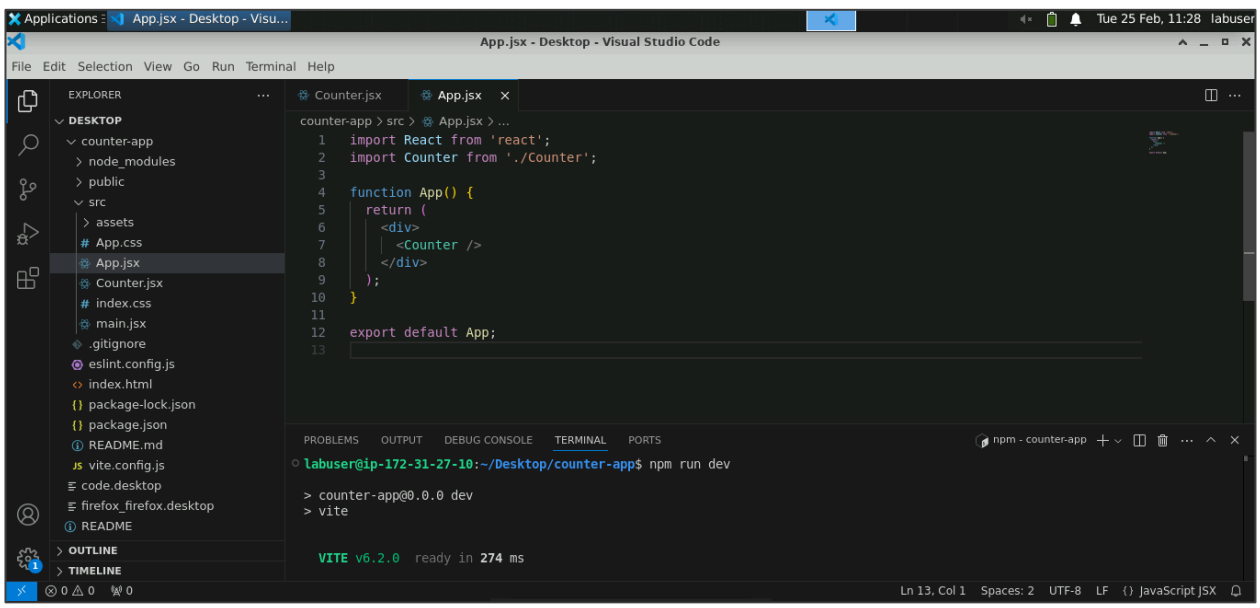
Step 3: Import and use the component

3.1 Open the **App.jsx** file inside the **src** folder



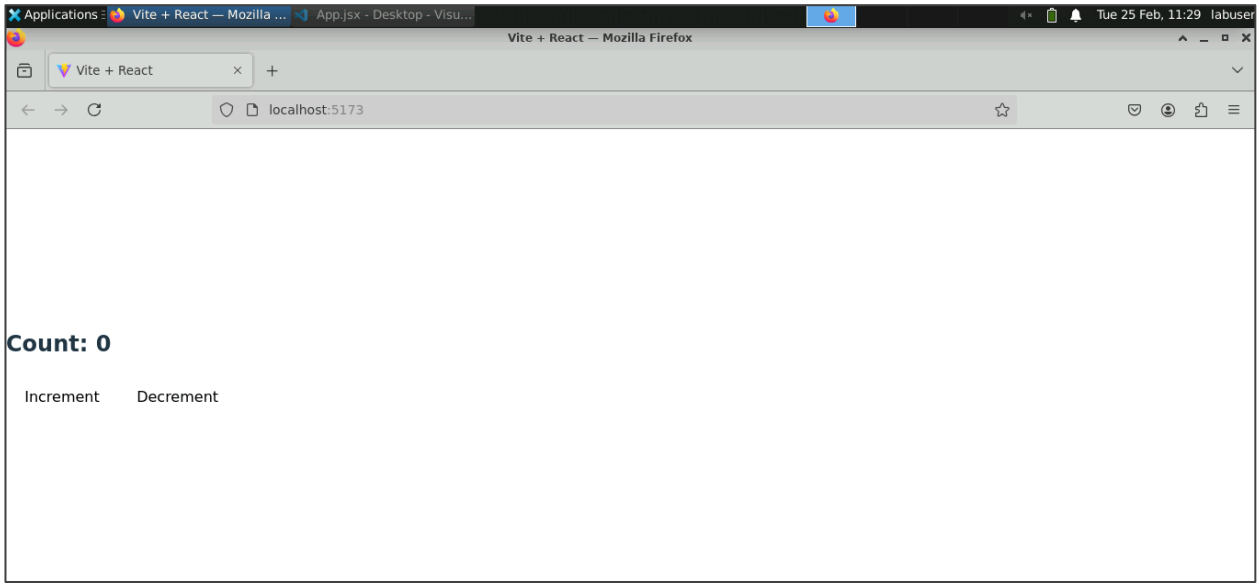
3.2 Paste the code below to include the new component in **App.jsx**:

```
import React from 'react';
import Counter from './Counter';

function App() {
  return (
    <div>
      <Counter />
    </div>
  );
}

export default App;
```

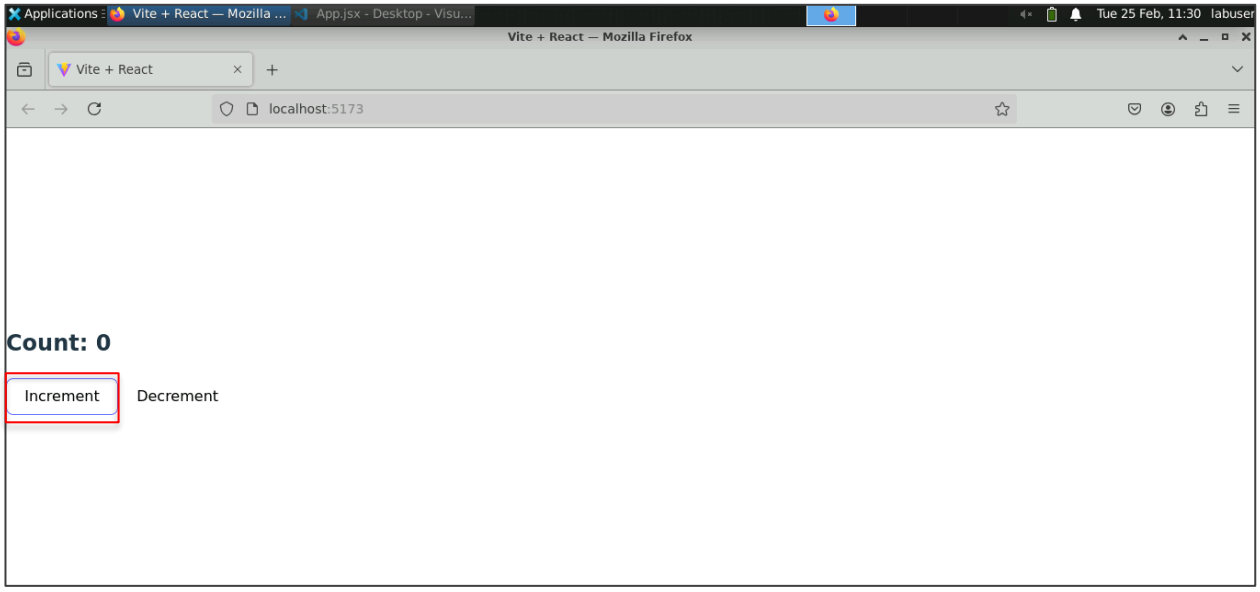


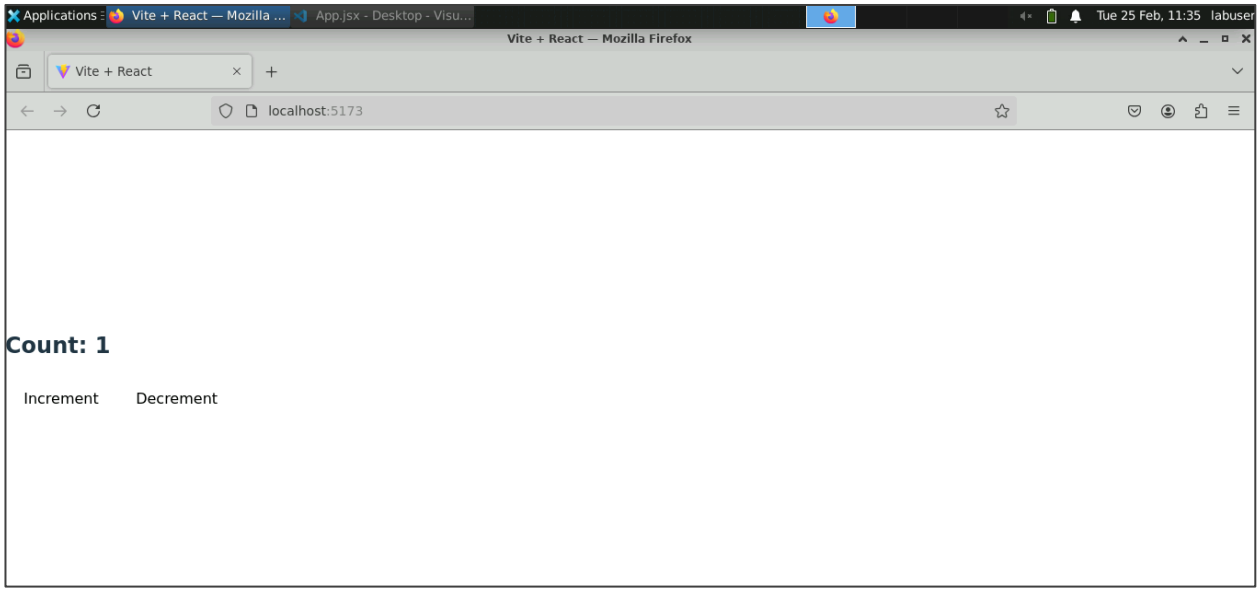
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage

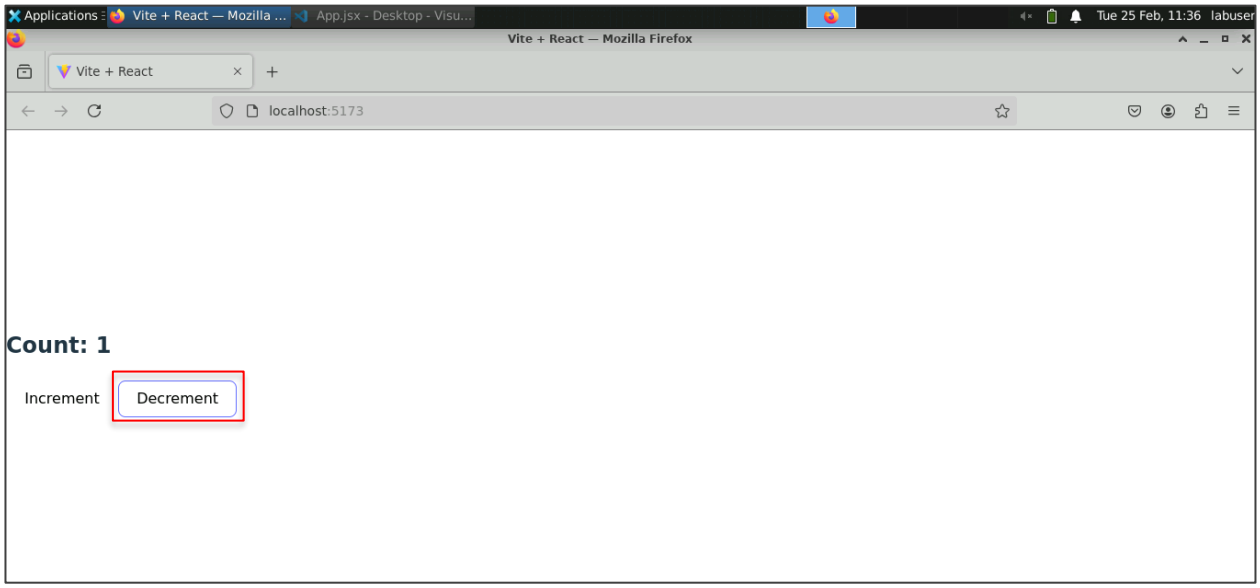


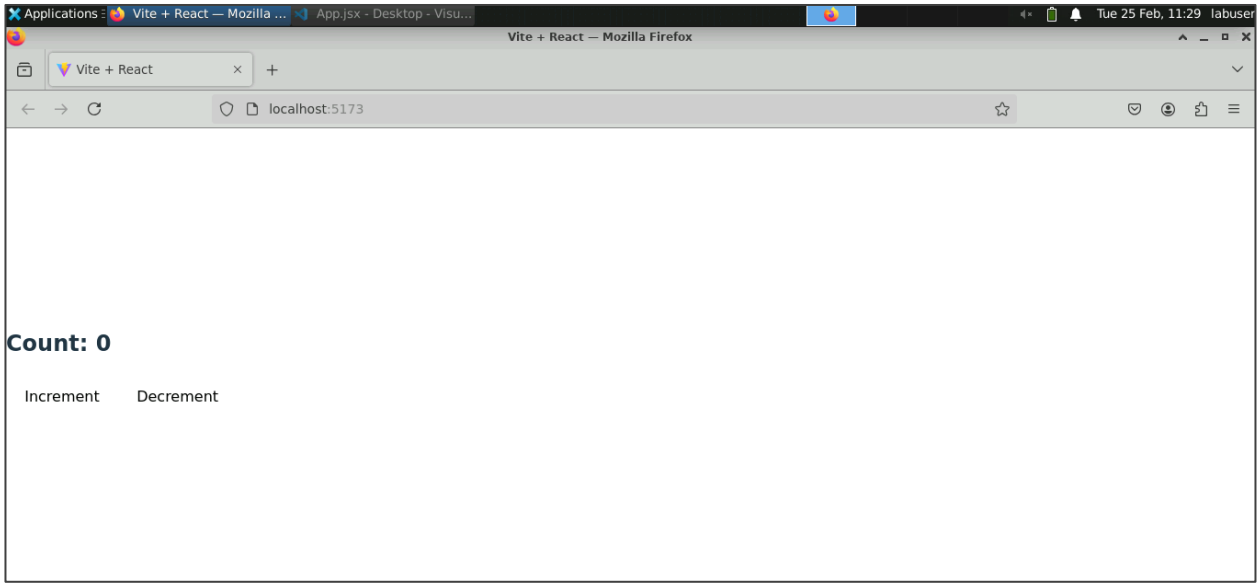
4.2 Click **Increment** to increment the count by 1





4.3 Click **Decrement** to decrement the count by 1





By following the above steps, you have successfully created a react component using the useState hook. This implementation allows the component to maintain and update state dynamically in response to user interactions.